

HW2

Sam Olson

Q1

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

Note: My answers rely on the row-space criterion for estimability, taking advantage of the following:

- **Theorem 2.1:** Under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' = \underline{a}'X$ for some $n \times 1$ vector $\underline{a}$; said differently, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R}(X)$, where $\mathcal{R}(X) = \{\underline{a}'X : \underline{a} \in \mathbb{R}^n\}$.
- **Lemma 2.1:** Under Models II–VII, a linear estimator $c + \underline{a}'\underline{y}$ of $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$ (used in Q4 and Q5; sorry, this is an artefact that is no longer used in Q1).

(a)

If $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are both estimable, is $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ necessarily estimable? Explain your answer.

Answer. Yes.

Since $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are estimable, Theorem 2.1 gives $n \times 1$ vectors $\underline{a}_1$ and $\underline{a}_2$ with $\underline{\lambda}'_1 = \underline{a}'_1X$ and $\underline{\lambda}'_2 = \underline{a}'_2X$. Then

$$(\underline{\lambda}_1 + \underline{\lambda}_2)' = (\underline{a}_1 + \underline{a}_2)'X \in \mathcal{R}(X),$$

so $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta} = (\underline{\lambda}_1 + \underline{\lambda}_2)'\underline{\beta}$ is estimable by Theorem 2.1.

(Said perhaps more succinctly: If both $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are estimable, then $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ is itself a linear combination of two estimable quantities, and is hence estimable.)

(b)

If $\underline{\lambda}'_1\underline{\beta}$ and $\underline{\lambda}'_2\underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ necessarily nonestimable? Explain your answer.

Answer. No.

Let $\underline{\lambda}'_1\underline{\beta}$ be nonestimable, so $\underline{\lambda}'_1 \notin \mathcal{R}(X)$ by Theorem 2.1, and set $\underline{\lambda}_2 = -\underline{\lambda}_1$. Since $\mathcal{R}(X)$ is a subspace, $\underline{\lambda}'_2 = -\underline{\lambda}'_1 \notin \mathcal{R}(X)$ as well, so $\underline{\lambda}'_2\underline{\beta}$ is also nonestimable.

However, by adding a “clever zero”,

$$\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta} = \underline{\lambda}'_1\underline{\beta} + (-\underline{\lambda}'_1\underline{\beta}) = \underline{0}'_p\underline{\beta} = 0,$$

which is estimable, as $\underline{0}'_p = \underline{0}'_nX \in \mathcal{R}(X)$.

Hence the sum of two nonestimable functions can be estimable, so $\underline{\lambda}'_1\underline{\beta} + \underline{\lambda}'_2\underline{\beta}$ is not necessarily nonestimable.

Note: This answer requires $\text{rank}(X) < p$; otherwise $\mathcal{R}(X) = \mathbb{R}^p$ and every $\underline{\lambda}'\underline{\beta}$ is estimable.

(c)

Can $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ be estimable if $\underline{\lambda}'_1\beta$ is estimable and $\underline{\lambda}'_2\beta$ is nonestimable? Explain your answer.

Answer. No ($\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ would necessarily be nonestimable).

Suppose, for contradiction, that $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ is estimable. By Theorem 2.1, $\underline{\lambda}'_1 + \underline{\lambda}'_2 \in \mathcal{R}(X)$ and $\underline{\lambda}'_1 \in \mathcal{R}(X)$, and since $\mathcal{R}(X)$ is a subspace,

$$\underline{\lambda}'_2 = (\underline{\lambda}'_1 + \underline{\lambda}'_2) - \underline{\lambda}'_1 \in \mathcal{R}(X).$$

So $\underline{\lambda}'_2\beta$ is estimable by Theorem 2.1, contradicting the assumption that it is nonestimable.

Q2

Take the model to be Model VII and suppose that

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a)

Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.

Answer. The projections are $(2, 3, 0)'$ on $\mathcal{C}(X)$ and $(0, 0, 4)'$ on $\mathcal{C}^\perp(X)$.

To find these projections, we utilize the following results/definitions:

- $\mathcal{C}(X) = \{X\underline{b} : \underline{b} \in \mathbb{R}^3\}$ is the span of the columns of X ,
- $\mathcal{C}^\perp(X) = \{\underline{a} \in \mathbb{R}^3 : X'\underline{a} = \underline{0}\}$ is the null space of X' , and
- **Theorem 2.6:** Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition $\underline{y} = \underline{y}_1 + \underline{y}_2$ with $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$.
 - What's more, this decomposition is given by $\underline{y}_1 = X\hat{\underline{\beta}}$ and $\underline{y}_2 = \underline{y} - X\hat{\underline{\beta}}$ for any solution $\hat{\underline{\beta}}$ to the normal equations.
 - And the vectors $\underline{y}_1$ and $\underline{y}_2$ are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

We'll next detail the computation of the projections.

$\mathcal{C}(X)$

The columns of X as given are

$$\underline{x}_1 = (1, 2, 0)', \quad \underline{x}_2 = (2, 3, 0)', \quad \underline{x}_3 = (3, 1, 0)'.$$

Each has a shared third coordinate, 0, such that $\mathcal{C}(X) \subseteq \{(u, v, 0)' : u, v \in \mathbb{R}\}$.

Also note that the vectors

$$-3\underline{x}_1 + 2\underline{x}_2 = (1, 0, 0)' \quad \text{and} \quad 2\underline{x}_1 - \underline{x}_2 = (0, 1, 0)'$$

both lie in $\mathcal{C}(X)$, and every $(u, v, 0)'$ is a linear combination of these two.

Hence

$$\mathcal{C}(X) = \{(u, v, 0)' : u, v \in \mathbb{R}\}.$$

$\mathcal{C}^\perp(X)$

For $\underline{a} = (a_1, a_2, a_3)'$,

$$X'\underline{a} = (a_1 + 2a_2, 2a_1 + 3a_2, 3a_1 + a_2)'.$$

Since $\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = -1 \neq 0$, the first two coordinates vanish only if $a_1 = a_2 = 0$. The third then vanishes as well, and a_3 is unrestricted. Hence

$$\mathcal{C}^\perp(X) = \{(0, 0, w)' : w \in \mathbb{R}\}.$$

Combining it all together.

Splitting $\underline{y}$,

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

with the first vector in $\mathcal{C}(X)$ and the second in $\mathcal{C}^\perp(X)$.

By the uniqueness in Theorem 2.6, these are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, resp.

(b)

Determine $\text{rank}(X'X)$.

Answer. $\text{rank}(X'X) = 2$.

Note: By **Lemma 2.3**, $\text{rank}(A'A) = \text{rank}(A)$ for any matrix A .

From part (a), $\mathcal{C}(X)$ is spanned by the two linearly independent vectors $(1, 0, 0)'$ and $(0, 1, 0)'$, so $\text{rank}(X) = \dim \mathcal{C}(X) = 2$.

Via Lemma 2.3, with $A = X$, we then have

$$\text{rank}(X'X) = \text{rank}(X) = 2.$$

(c)

Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

Answer.

By definition, a *generalized inverse* of a matrix A is any matrix G satisfying $AGA = A$.

Here, we have:

$$A = X'X = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix},$$

which has rank $r = 2$ by part (b).

Then

$$B = \begin{pmatrix} 5 & 8 \\ 8 & 13 \end{pmatrix}, \quad \det(B) = 65 - 64 = 1 \neq 0, \quad (B^{-1})' = B^{-1} = \begin{pmatrix} 13 & -8 \\ -8 & 5 \end{pmatrix}.$$

Utilizing the method noted (replacing the entries of B in A by those of $(B^{-1})'$, and all other entries by 0), we have

$$G = (X'X)^- = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

And since B is symmetric, the resulting matrix is symmetric, so its transpose (the final step of the method) is equal to itself.

We then verify this is in fact a generalized inverse:

$$AG = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 5 & 0 \end{pmatrix} \Rightarrow AGA = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix} = A.$$

So G is a generalized inverse of $X'X$, as desired.

(d)

Compute a solution $\hat{\underline{\beta}}$ to the normal equations.

Answer. $\hat{\underline{\beta}} = (0, 1, 0)'$.

To compute this solution, we utilize the following results:

- **Theorem 2.3:** The normal equations $X'X\hat{\underline{\beta}} = X'\underline{y}$ are consistent.
- **Theorem 1.2:** A matrix G is a generalized inverse of a matrix A if and only if $\underline{w} = G\underline{c}$ solves $A\underline{w} = \underline{c}$ for every vector $\underline{c}$ for which these equations are consistent.

With these results in hand, since the normal equations are consistent (Theorem 2.3), Theorem 1.2 with $A = X'X$ and $\underline{c} = X'\underline{y}$ gives that $\hat{\underline{\beta}} = (X'X)^- X'\underline{y}$ solves them.

Since

$$X'\underline{y} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix},$$

taking $(X'X)^- = G$ from part (c) gives

$$\hat{\underline{\beta}} = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix} = \begin{pmatrix} 104 - 104 \\ -64 + 65 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

(e)

Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

Answer.

From part (d), $X\hat{\underline{\beta}}$ is the second column of X , so

$$X\hat{\underline{\beta}} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \underline{y} - X\hat{\underline{\beta}} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix},$$

which match the projections in part (a), as Theorem 2.6 requires.

A quick numerical check of parts (c)–(e) in R:

```
X <- matrix(c(1, 2, 3,
              2, 3, 1,
              0, 0, 0),
            nrow = 3,
            byrow = TRUE)
y <- c(2, 3, 4)
XtX <- crossprod(X)
G <- matrix(c(13, -8, 0,
              -8, 5, 0,
              0, 0, 0),
            nrow = 3,
            byrow = TRUE)

# verify G is a generalized inverse of X'X
XtX %*% G %*% XtX

##      [,1] [,2] [,3]
## [1,]    5    8    5
## [2,]    8   13    9
## [3,]    5    9   10

XtX

##      [,1] [,2] [,3]
## [1,]    5    8    5
## [2,]    8   13    9
## [3,]    5    9   10

bhat <- G %*% crossprod(X, y)
# bhat
# Expect (0, 1, 0)
bhat

##      [,1]
## [1,]    0
## [2,]    1
## [3,]    0

# X bhat
# Expect (2, 3, 0)
X %*% bhat

##      [,1]
## [1,]    2
## [2,]    3
## [3,]    0

# y - X bhat
# Expect (0, 0, 4)
y - X %*% bhat
```

```
##      [,1]
## [1,]    0
## [2,]    0
## [3,]    4
```

Q3

For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{0'\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is the same as the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}'X'y = \underline{a}'X'_1y_1 + \underline{a}'X'_2y_2$ and, under the partial model, the LSE is $\underline{b}'X'_1y_1$ where here $X'_1X_1\underline{a} + X'_2X_2\underline{a} = X'X\underline{a} = \underline{\lambda} = X'_1X_1\underline{b}$ (why?); in the last equation, move terms involving X'_2 and X'_1 to different sides.

Answer.

Note: X_1 is $n_1 \times p$ and X_2 is $n_2 \times p$, with $n_1 + n_2 = n$. The partial model $y_1 = X_1\beta + \epsilon_1$ is itself an instance of Model VII, as $E(\underline{\epsilon}_1) = \underline{0}_{n_1}$ and $\text{Var}(\underline{\epsilon}_1)$ is the leading $n_1 \times n_1$ block of $\text{Var}(\underline{\epsilon})$. This means we can apply key (known) results to both models based on X_1 and X , resp., in this problem.

That said, the key results we will use are as follows:

- **Theorem 2.5:** Under Models II–VII, $\underline{\lambda}'\underline{\hat{\beta}} = \underline{\lambda}'\underline{\tilde{\beta}}$ for any two solutions $\underline{\hat{\beta}}$ and $\underline{\tilde{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.
- **Theorem 2.7:** Under Models II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the conjugate normal equations $X'X\underline{r} = \underline{\lambda}$ are consistent.
- **Theorem 2.8:** Under Models II–VII, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ equals $\underline{r}'X'y$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.
- **Lemma 1.2:** For any matrix M , $\text{tr}(M'M) = 0$ implies $M = 0$.
- **Lemma 1.3(ii):** For any matrices M , P , and Q of appropriate dimensions, $M'MP = M'MQ$ implies $MP = MQ$.

On Estimability.

Since $\underline{\lambda}' \in \mathcal{R}(X_1)$, we have $\underline{\lambda}' = \underline{t}'X_1$ for some $n_1 \times 1$ vector $\underline{t}$, so $\underline{\lambda}'\underline{\beta}$ is estimable under the partial model by Theorem 2.1 (noted in Q1).

Also, $\underline{\lambda}'\underline{\beta}$ is estimable as well under the full model, since

$$\underline{\lambda}' = (\underline{t}', \underline{0}'_{n_2}) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in \mathcal{R}(X).$$

By Theorem 2.5, its least squares estimate is then well defined under *each* model. The two models have different normal equations, though, so it remains to show that the two estimates coincide.

The two estimates (and utilizing the hint).

Since $\underline{\lambda}'\underline{\beta}$ is estimable under both models, Theorem 2.7 gives $p \times 1$ vectors $\underline{r}$ and $\underline{r}_1$ (the hint's $\underline{a}$ and $\underline{b}$) with

$$X'X\underline{r} = \underline{\lambda} \quad \text{and} \quad X'_1X_1\underline{r}_1 = \underline{\lambda},$$

and by Theorem 2.8, the least squares estimates under the full and partial models are $\underline{r}'X'y$ and $\underline{r}'_1X'_1y_1$, resp.

By block multiplication,

$$X'X = X_1'X_1 + X_2'X_2 \quad \text{and} \quad X'y = X_1'y_1 + X_2'y_2,$$

so

$$X_1'X_1\bar{r} + X_2'X_2\bar{r} = \bar{\lambda} = X_1'X_1\bar{r}_1, \quad \text{that is,} \quad X_1'X_1(\bar{r} - \bar{r}_1) = -X_2'X_2\bar{r}.$$

Then, transposing both sides gives an equality of $1 \times p$ row vectors,

$$\underbrace{[X_1(\bar{r} - \bar{r}_1)]'}_{\in \mathcal{R}(X_1)} X_1 = \underbrace{[-X_2\bar{r}']}_{\in \mathcal{R}(X_2)} X_2.$$

The left side is $\bar{u}'X_1$ with $\bar{u} = X_1(\bar{r} - \bar{r}_1) \in \mathbb{R}^{n_1}$, so it lies in $\mathcal{R}(X_1)$; similarly, the right side lies in $\mathcal{R}(X_2)$. Since $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{\bar{0}'\}$, both sides equal $\bar{0}'_p$.

This gives two facts:

1. $\bar{r}'X_2'X_2 = \bar{0}'_p$, so $(X_2\bar{r})'(X_2\bar{r}) = \bar{r}'X_2'X_2\bar{r} = 0$. Then, by Lemma 1.2 with $M = X_2\bar{r}$ (an $n_2 \times 1$ matrix, for which $\text{tr}(M'M) = M'M = 0$), $X_2\bar{r} = \bar{0}_{n_2}$.
2. $X_1'X_1\bar{r} = X_1'X_1\bar{r}_1$. Then, by Lemma 1.3(ii) with $M = X_1$, $P = \bar{r}$, and $Q = \bar{r}_1$, $X_1\bar{r} = X_1\bar{r}_1$.

Combining the two facts noted above,

$$\bar{r}'X'y = \bar{r}'X_1'y_1 + \bar{r}'X_2'y_2 = (X_1\bar{r})'y_1 + (X_2\bar{r})'y_2 = (X_1\bar{r}_1)'y_1 + 0 = \bar{r}_1'X_1'y_1,$$

so the least squares estimates of $\bar{\lambda}'\bar{\beta}$ under the two models are one and the same.

Q4

Show that, under Model II, $\bar{a}'\bar{y}$ is a BLUE of its expectation if and only if $\bar{a} \in \mathcal{C}(X)$.

Answer.

Note: Under Model II, we have:

- $\bar{a}$ is $n \times 1$,
- $E(\bar{\epsilon}) = \bar{0}_n$, and
- $\text{Var}(\bar{y}) = \text{Var}(\bar{\epsilon}) = \sigma^2 I_n$ with $\sigma^2 > 0$.

We also utilize the following results:

- **Lemma 1.4:** For any matrix X , $X'X(X'X)^-X' = X'$.
- **Corollary 1(ii) to Lemma 1.4:** For any matrix X , $X(X'X)^-X'X = X$.
- **Corollary 3 to Lemma 1.5:** For any matrix X , $X(X'X)^-X'$ is symmetric.
- **Corollary to Lemma 2.4:** For a random vector $\bar{x}$ with $\text{Var}(\bar{x}) = \Sigma$ and a matrix of constants L , $\text{Var}(L'\bar{x}) = L'\Sigma L$.
- **Theorem 2.9 (Gauss–Markov):** Under Models II–VII, if $\bar{\lambda}'\bar{\beta}$ is estimable and $\widehat{\bar{\beta}}$ is any solution to the normal equations, then $\bar{\lambda}'\widehat{\bar{\beta}}$ is a linear unbiased estimator of $\bar{\lambda}'\bar{\beta}$. What's more, under Model II, the variance of $\bar{\lambda}'\widehat{\bar{\beta}}$ is no greater than that of any other linear unbiased estimator of $\bar{\lambda}'\bar{\beta}$, for every $\sigma^2 > 0$; said differently, $\bar{\lambda}'\widehat{\bar{\beta}}$ is the BLUE.

We also utilize the properties of the projection matrix, $P_X = X(X'X)^-X'$, an $n \times n$ matrix that satisfies:

- $P_X X = X$,
- $P_X' = P_X$, and
- $P_X^2 = X(X'X)^- [X'X(X'X)^-X'] = X(X'X)^-X' = P_X$.

It then follows that $P'_X(I_n - P_X) = P_X - P_X^2 = 0_{n \times n}$.

Taking this information as given, then, we can proceed with the proof.

Proof:

We have $E(\underline{a}'\underline{y}) = \underline{a}'X\underline{\beta} = \underline{\lambda}'\underline{\beta}$ with $\underline{\lambda}' = \underline{a}'X \in \mathcal{R}(X)$.

By Lemma 2.1 (noted in Q1), $\underline{a}'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and so $\underline{\lambda}'\underline{\beta}$ is estimable.

By Theorems 2.3 and 1.2 (noted in Q2(d)), $\widehat{\underline{\beta}} = (X'X)^{-}X'\underline{y}$ solves the normal equations, so by Theorem 2.9, the BLUE of $\underline{\lambda}'\underline{\beta}$ is

$$\underline{\lambda}'\widehat{\underline{\beta}} = \underline{a}'X(X'X)^{-}X'\underline{y} = \underline{a}'P_X\underline{y} = (P_X\underline{a})'\underline{y},$$

where the last step uses $P'_X = P_X$.

Comparing variances.

Applying the Corollary to Lemma 2.4 with $L = \underline{a}$ and $\Sigma = \sigma^2 I_n$, and writing $\underline{a} = P_X\underline{a} + (I_n - P_X)\underline{a}$,

$$\text{Var}(\underline{a}'\underline{y}) = \sigma^2 \underline{a}'\underline{a} = \underbrace{\sigma^2 (P_X\underline{a})'(P_X\underline{a})}_{(1)} + \underbrace{2\sigma^2 \underline{a}'P'_X(I_n - P_X)\underline{a}}_{(2)} + \underbrace{\sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}]}_{(3)}.$$

The first term (1) is $\text{Var}((P_X\underline{a})'\underline{y})$, by the Corollary to Lemma 2.4.

And the middle term (2) vanishes because $P'_X(I_n - P_X) = 0_{n \times n}$.

Hence

$$\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y}) + \sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}]. \quad (\text{Q4.1})$$

Then, since $\underline{a}'\underline{y}$ is linear and unbiased for $\underline{\lambda}'\underline{\beta}$, and $(P_X\underline{a})'\underline{y}$ is the BLUE, $\underline{a}'\underline{y}$ is a BLUE if and only if $\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y})$.

Because $\sigma^2 > 0$, this holds if and only if $[(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}] = 0$.

By Lemma 1.2 (noted in Q3), applied to the $n \times 1$ matrix $(I_n - P_X)\underline{a}$, this holds if and only if $(I_n - P_X)\underline{a} = \underline{0}_n$, that is, $\underline{a} = P_X\underline{a}$.

To finalize the proof, it remains to show $\underline{a} = P_X\underline{a}$ if and only if $\underline{a} \in \mathcal{C}(X)$, that is, $\underline{a} = X\underline{b}$ for some $p \times 1$ vector $\underline{b}$:

- ($\Rightarrow$) If $\underline{a} = P_X\underline{a}$, then $\underline{a} = X\underline{b}$ with $\underline{b} = (X'X)^{-}X'\underline{a}$, so $\underline{a} \in \mathcal{C}(X)$.
- ($\Leftarrow$) If $\underline{a} = X\underline{b}$, then $P_X\underline{a} = P_XX\underline{b} = X\underline{b} = \underline{a}$, since $P_XX = X$.

Hence $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Q5

(a)

Show that, under Model II, the problem of determining the BLUE $\underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is equivalent to that of minimizing $\underline{a}'\underline{a}$ subject to the constraints $\underline{a}'X = \underline{\lambda}'$.

Answer.

Note: $\underline{a}$ is $n \times 1$ and $\underline{\lambda}$ is $p \times 1$.

By Lemma 2.1 (noted in Q1), a linear estimator $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ if and only if $c = 0$ and $\underline{a}'X = \underline{\lambda}'$.

So the linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$ are exactly

$$\{\underline{a}'\underline{y} : \underline{a}'X = \underline{\lambda}'\},$$

which is nonempty because $\underline{\lambda}'\underline{\beta}$ is estimable.

Under Model II, $\text{Var}(\underline{y}) = \sigma^2 I_n$, so by the Corollary to Lemma 2.4 (noted in Q4), each has variance

$$\text{Var}(\underline{a}'\underline{y}) = \underline{a}'(\sigma^2 I_n)\underline{a} = \sigma^2 \underline{a}'\underline{a}.$$

Since $\sigma^2 > 0$ does not depend on $\underline{a}$, it then follows that $\underline{a}$ minimizes $\text{Var}(\underline{a}'\underline{y})$ over this class if and only if it minimizes $\underline{a}'\underline{a}$ subject to $\underline{a}'X = \underline{\lambda}'$ (equivalently, $X'\underline{a} = \underline{\lambda}$).

And since the minimizing $\underline{a}$ does not depend on σ^2 , the resulting estimator has minimum variance uniformly in σ^2 .

(b)

The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

Answer.

Note: There is one multiplier per constraint in $X'\underline{a} = \underline{\lambda}$, so $\underline{\rho}$ is $p \times 1$.

Also note the following identities for a vector variable $\underline{\beta}$:

- For any constant vector $\underline{h}$, $\partial(\underline{h}'\underline{\beta})/\partial\underline{\beta} = \underline{h}$.
- For any constant matrix X , $\partial(\underline{\beta}'X'X\underline{\beta})/\partial\underline{\beta} = 2X'X\underline{\beta}$.

Applying these term by term:

- *With respect to $\underline{a}$.* The second identity, with I_n in place of X and $\underline{a}$ in place of $\underline{\beta}$, gives $\partial(\underline{a}'\underline{a})/\partial\underline{a} = 2\underline{a}$. Writing $\underline{\rho}'X'\underline{a} = (X\underline{\rho})'\underline{a}$, the first identity with $\underline{h} = X\underline{\rho}$ gives $\partial(\underline{\rho}'X'\underline{a})/\partial\underline{a} = X\underline{\rho}$. The term $-2\underline{\rho}'\underline{\lambda}$ does not involve $\underline{a}$.
- *With respect to $\underline{\rho}$.* The term $\underline{a}'\underline{a}$ does not involve $\underline{\rho}$. Writing $2\underline{\rho}'(X'\underline{a} - \underline{\lambda}) = 2(X'\underline{a} - \underline{\lambda})'\underline{\rho}$, the first identity with $\underline{h} = X'\underline{a} - \underline{\lambda}$ applies.

Together, the $n \times 1$ and $p \times 1$ gradients are

$$\frac{\partial L}{\partial \underline{a}} = 2\underline{a} + 2X\underline{\rho}, \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X'\underline{a} - \underline{\lambda}).$$

Setting these equal to zero gives the system

$$\underline{a} + X\underline{\rho} = \underline{0}_n, \quad X'\underline{a} = \underline{\lambda}. \quad (\text{Q5 (b)})$$

(c)

Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.

Answer.

($\Rightarrow$) Suppose $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5 (b)). The first equation gives $\underline{a}_0 = -X\underline{\rho}_0$, and substituting into the second,

$$\underline{\lambda} = X'\underline{a}_0 = X'(-X\underline{\rho}_0).$$

So $\underline{r} = -\underline{\rho}_0$ solves $X'X\underline{r} = \underline{\lambda}$, and the conjugate normal equations are consistent.

($\Leftarrow$) Suppose $X'X\underline{r}_0 = \underline{\lambda}$ for some $p \times 1$ vector $\underline{r}_0$. Set $\underline{a}_0 = X\underline{r}_0$ and $\underline{\rho}_0 = -\underline{r}_0$. Then

$$\underline{a}_0 + X\underline{\rho}_0 = X\underline{r}_0 - X\underline{r}_0 = \underline{0}_n \quad \text{and} \quad X'\underline{a}_0 = X'X\underline{r}_0 = \underline{\lambda},$$

so $(\underline{a}_0, \underline{\rho}_0)$ solves (Q5 (b)).

(d)

Solve your equations for the minimizing value of $\underline{a}$.

Answer. $\underline{a}_* = X(X'X)^{-}\underline{\lambda}$.

We utilize the following results in this proof:

- **Theorem 1.3:** All solutions to a consistent matrix equation $AW = Y$ are generated by $W = A^{-}Y + (I - A^{-}A)Z$ as Z ranges over all matrices of appropriate dimensions. (We write W for the notes' unknown to avoid a clash with the design matrix X .)
- **Corollary to Lemma 1.1:** For any matrix X , $[(X'X)^{-}]'$ is a generalized inverse of $X'X$.

A solution exists.

Since $\underline{\lambda}'\underline{\beta}$ is estimable, the conjugate normal equations are consistent by Theorem 2.7 (noted in Q3), so (Q5(b)) has a solution by part (c).

All solutions.

The first equation of (Q5 (b)) gives $\underline{a} = -X\underline{\rho}$. Substituting into the second gives $X'X(-\underline{\rho}) = \underline{\lambda}$, a consistent system in the $p \times 1$ unknown $-\underline{\rho}$. Applying Theorem 1.3 with $A = X'X$, $W = -\underline{\rho}$, and $\bar{Y} = \underline{\lambda}$, all solutions are

$$-\underline{\rho} = (X'X)^{-}\underline{\lambda} + [I_p - (X'X)^{-}X'X]\underline{z}, \quad \underline{z} \in \mathbb{R}^p.$$

Hence

$$\underline{a} = -X\underline{\rho} = X(X'X)^{-}\underline{\lambda} + [X - X(X'X)^{-}X'X]\underline{z} = X(X'X)^{-}\underline{\lambda},$$

since $X(X'X)^{-}X'X = X$ by Corollary 1(ii) to Lemma 1.4 (noted in Q4). So every solution of (Q5 (b)) has the same $\underline{a}$ -component,

$$\underline{a}_* = X(X'X)^{-}\underline{\lambda}.$$

Then note that equations (Q5 (b)) are necessary conditions only, so we verify directly that $\underline{a}_*$ minimizes $\underline{a}'\underline{a}$ subject to $X'\underline{a} = \underline{\lambda}$. Let $\underline{a}$ be any $n \times 1$ vector with $X'\underline{a} = \underline{\lambda}$, and set $\underline{d} = \underline{a} - \underline{a}_*$, so that $X'\underline{d} = \underline{\lambda} - \underline{\lambda} = \underline{0}_p$. Since $\underline{a}_* = X\underline{r}_*$ with $\underline{r}_* = (X'X)^{-}\underline{\lambda}$,

$$\underline{a}'_*\underline{d} = \underline{r}'_*X'\underline{d} = 0,$$

and therefore

$$\underline{a}'\underline{a} = (\underline{a}_* + \underline{d})'(\underline{a}_* + \underline{d}) = \underline{a}'_*\underline{a}_* + \underline{d}'\underline{d} \geq \underline{a}'_*\underline{a}_*.$$

Equality holds if and only if $\underline{d}'\underline{d} = 0$, that is, $\underline{d} = \underline{0}_n$ (by Lemma 1.2, noted in Q3). So $\underline{a}_*$ is the unique minimizer.

Then, by part (a), the BLUE is

$$\underline{a}'_*\underline{y} = \underline{\lambda}'[(X'X)^{-}]'X'\underline{y}.$$

By the Corollary to Lemma 1.1, $[(X'X)^{-}]'$ is itself a generalized inverse of $X'X$, so by Theorems 2.3 and 1.2 (noted in Q2(d)), $\widehat{\underline{\beta}} = [(X'X)^{-}]'X'\underline{y}$ solves the normal equations, and

$$\underline{a}'_*\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}}.$$

That is, the BLUE of $\underline{\lambda}'\underline{\beta}$ is its least squares estimator.